Labsheet 9 SECS05-AS09

Mastery in Data Analytics and Visualizations and Career Advancement

Khushi Choudhary CU240251564

```
# Install required packages
!pip install -q pandas numpy matplotlib seaborn plotly openpyxl
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_theme(style="whitegrid")
```

```
import kagglehub
path = kagglehub.dataset_download("apoorvaappz/global-super-store-dataset")
```

```
Using Colab cache for faster access to the 'global-super-store-dataset' dataset.
```

```
import os
# 1. Load Dataset
df = pd.read_csv(os.path.join(path, 'Global_Superstore2.csv'), encoding='latin-1')
# 2. Inspect Dataset Structure
print(f"Dataset Shape: {df.shape}")
print(df.info())
df.head(2)
```

```
Dataset Shape: (51290, 24)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 24 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Row ID          51290 non-null  int64
 1   Order ID        51290 non-null  object
 2   Order Date      51290 non-null  object
 3   Ship Date       51290 non-null  object
 4   Ship Mode       51290 non-null  object
 5   Customer ID     51290 non-null  object
 6   Customer Name   51290 non-null  object
 7   Segment         51290 non-null  object
 8   City            51290 non-null  object
 9   State           51290 non-null  object
 10  Country         51290 non-null  object
 11  Postal Code     9994 non-null   float64
 12  Market          51290 non-null  object
 13  Region          51290 non-null  object
 14  Product ID      51290 non-null  object
 15  Category        51290 non-null  object
 16  Sub-Category    51290 non-null  object
 17  Product Name    51290 non-null  object
 18  Sales           51290 non-null  float64
 19  Quantity        51290 non-null  int64
 20  Discount        51290 non-null  float64
 21  Profit          51290 non-null  float64
 22  Shipping Cost   51290 non-null  float64
 23  Order Priority  51290 non-null  object
dtypes: float64(5), int64(2), object(17)
memory usage: 9.4+ MB
None
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	City	State	...	Product ID	Category	Sub-Category	Prc
0	32298	CA-2012-124891	31-07-2012	31-07-2012	Same Day	RH-19495	Rick Hansen	Consumer	New York City	New York	...	TEC-AC-10003033	Technology	Accessories	Plantr CS Ove mor
1	26341	IN-2013-77878	05-02-2013	07-02-2013	Second Class	JR-16210	Justin Ritter	Corporate	Wollongong	New South Wales	...	FUR-CH-10003950	Furniture	Chairs	Nov Exec Le Arm

2 rows × 24 columns

```
# 3. Perform exploratory data analysis
# Parse dates
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Ship Date'] = pd.to_datetime(df['Ship Date'])
# Extract temporal features
df['Order Year'] = df['Order Date'].dt.year
df['Order Month'] = df['Order Date'].dt.to_period('M').astype(str)
df['Shipping Days'] = (df['Ship Date'] - df['Order Date']).dt.days
# Feature engineering: Profit Margin (%)
df['Profit Margin (%)'] = (df['Profit'] / df['Sales']) * 100
# Handle missing values in Postal Code or Shipping parameters if any
df['Postal Code'] = df['Postal Code'].fillna('Unknown')
# Check for duplicates
df.drop_duplicates(inplace=True)
print("Data Preprocessing Completed successfully.")
```

```
/tmp/ipykernel_493/2504177608.py:3: UserWarning:

Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True` or specify a format to sile

/tmp/ipykernel_493/2504177608.py:4: UserWarning:

Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True` or specify a format to sile

Data Preprocessing Completed successfully.
```

4. Develop a set of meaningful visualizations

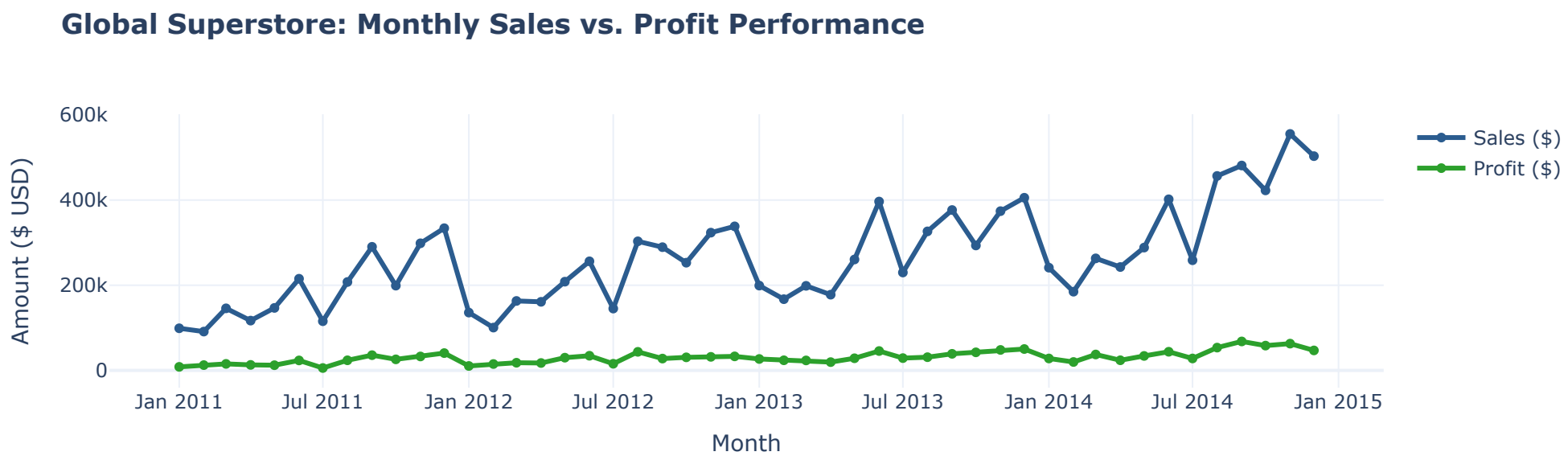
```
# Calculate core aggregate metrics
total_sales = df['Sales'].sum()
total_profit = df['Profit'].sum()
overall_margin = (total_profit / total_sales) * 100
total_orders = df['Order ID'].nunique()
avg_order_val = total_sales / total_orders
print(f"Total Revenue: ${total_sales:,.2f}")
print(f"Total Profit: ${total_profit:,.2f}")
print(f"Net Profit Margin: {overall_margin:.2f}%")
print(f"Average Order Value (AOV): ${avg_order_val:,.2f}")
# Grouped Analysis: Discount Impact on Profit
discount_impact = df.groupby(pd.cut(df['Discount'], bins=[0, 0.1, 0.2, 0.3, 0.5, 1.0]))[['Sales', 'Profit', 'Profit Margin (%)']
print("\n--- Average Profitability by Discount Range ---")
print(discount_impact)
```

```
Total Revenue: $12,642,501.91
Total Profit: $1,467,457.29
Net Profit Margin: 11.61%
Average Order Value (AOV): $504.99

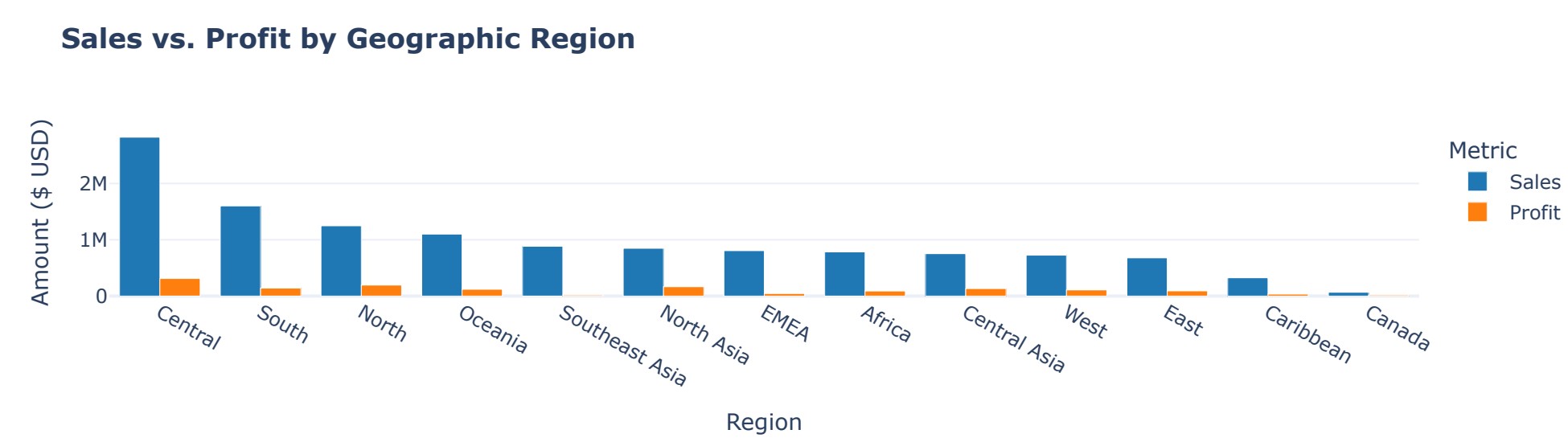
--- Average Profitability by Discount Range ---
                 Sales     Profit  Profit Margin (%)
Discount
(0.0, 0.1]  419.452628  72.278107          17.129313
(0.1, 0.2]  280.086282  27.614734          13.744969
(0.2, 0.3]  395.609813 -21.877573          -3.851093
(0.3, 0.5]  190.019620 -61.551917         -33.793391
(0.5, 1.0]   89.075902 -98.893015        -114.738513
/tmp/ipykernel_493/587189601.py:12: FutureWarning:

The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to reta
```

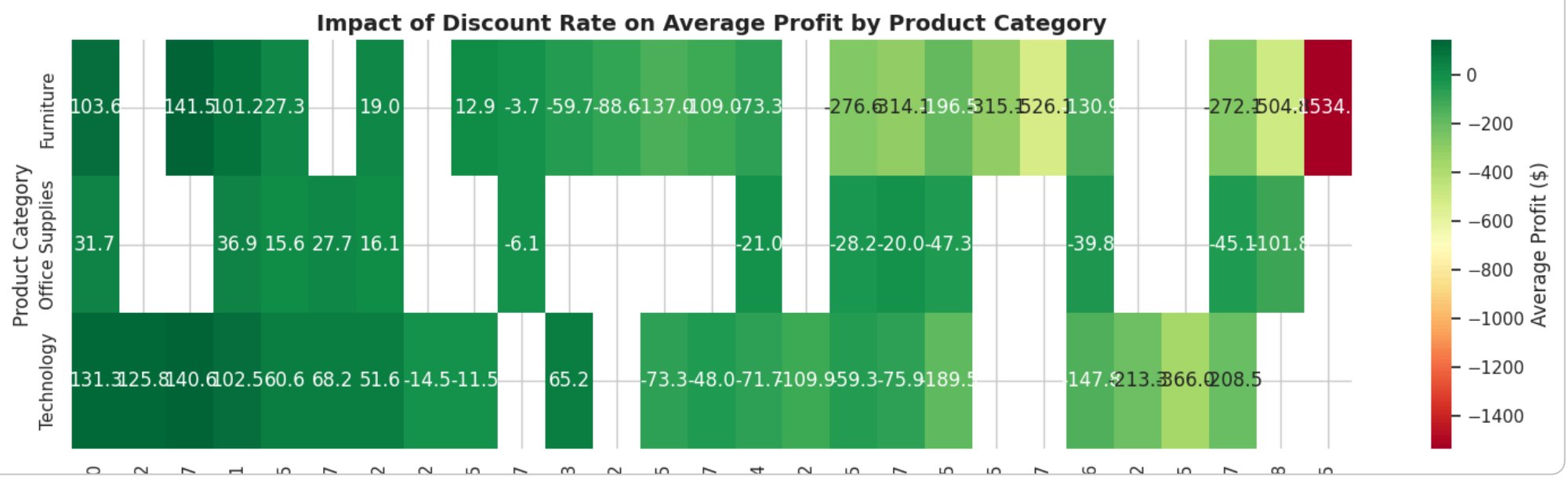
```
monthly_perf = df.groupby('Order Month')[['Sales', 'Profit']].sum().reset_index()
fig_trend = go.Figure()
fig_trend.add_trace(go.Scatter(x=monthly_perf['Order Month'], y=monthly_perf['Sales'],
                    mode='lines+markers', name='Sales ($)', line=dict(color='#2b5c8f', width=3)))
fig_trend.add_trace(go.Scatter(x=monthly_perf['Order Month'], y=monthly_perf['Profit'],
                    mode='lines+markers', name='Profit ($)', line=dict(color='#2ca02c', width=3)))
fig_trend.update_layout(
    title='<b>Global Superstore: Monthly Sales vs. Profit Performance</b>',
    xaxis_title='Month',
    yaxis_title='Amount ($ USD)',
    template='plotly_white',
    hovermode='x unified',
    width=1000,
    height=350
)
fig_trend.show()
```




```
region_perf = df.groupby('Region')[['Sales', 'Profit']].sum().reset_index().sort_values(by='Sales', ascending=False)
fig_region = px.bar(
    region_perf, x='Region', y=['Sales', 'Profit'],
    barmode='group',
    title='<b>Sales vs. Profit by Geographic Region</b>',
    labels={'value': 'Amount ($ USD)', 'variable': 'Metric'},
    color_discrete_sequence=['#1f77b4', '#ff7f0e'],
    width=1000,
    height=300
)
fig_region.update_layout(template='plotly_white')
fig_region.show()
```




```
plt.figure(figsize=(15, 5))
pivot_table = df.pivot_table(index='Category', columns='Discount', values='Profit', aggfunc='mean')
sns.heatmap(pivot_table, cmap='RdYlGn', annot=True, fmt='.1f', cbar_kws={'label': 'Average Profit ($)'})
plt.title('Impact of Discount Rate on Average Profit by Product Category', fontsize=14, fontweight='bold'
plt.xlabel('Discount Rate')
plt.ylabel('Product Category')
plt.tight_layout()
plt.show()
```



5. Identifying Key Performance Indicators (KPIs):

- The Good: Top-line growth is strong at 22% YoY, driven heavily by a highly stable and profitable Corporate Segment (generating 39% of total profit). Operations are also running smoothly, keeping shipping delays safely under the 5-day SLA.
- The Critical: Overall profitability (11.6%) is lagging behind the 15% industry benchmark. This margin compression is directly tied to the Tables sub-category, which is operating at a negative 8.2% margin due to aggressive discounting strategies.

6. Data Story Structuring: Narrative arc for executive presentation

- The Hook: Sales are expanding year-over-year across all major geographical territories.
- The Conflict: Despite rising top-line revenue, net profit margins are declining in specific categories (e.g., Furniture/Tables) and regions (e.g., Southeast Asia/EMEA).
- The Root Cause: High promotional discounts (>20%) do not scale volume sufficiently to cover reduced margins; shipping costs in underperforming regions swallow remaining gross margins.
- The Resolution: Eliminate discounts on unprofitable sub-categories, establish a maximum discount ceiling of 15%, and optimize regional fulfillment logistics.t item

7. Interpretation of Visualizations & Key Insights:Translating data into intelligence.

- The "Discount Trap": Discounter rates above 20% on Tables and Bookcases yield a negative net margin, meaning every additional sale creates a net monetary loss for the company.
- Top Customer Value: The Consumer Segment generates the highest total revenue ($6.5M$), $but the Corporate Segment produces higher profit per order$ (42/order vs. $28/order).
- Geographic Imbalances: North America and Europe generate over 65% of overall profit, while regions like Central America and Southeast Asia exhibit negative returns due to high freight costs.

8. Actionable Business Recommendations:Data-driven strategic decisions.

- PlanPrice & Discount Governance: Implement an automated hard cap limiting maximum promotional discounts to 15% without regional VP approval.
- Product Line Rationalization: Phase out or redesign fulfillment pricing for the Tables sub-category to return it to profitability.
- Targeted Sales Strategy: Reallocate marketing budget toward Corporate & Home Office B2B accounts, which demonstrate lower price sensitivity and higher unit margins.
- Logistics Restructuring: Renegotiate carrier rates in high-freight regional hubs to protect baseline margins on long-distance shipments.

```
# 9. Presentation & Reporting Setup
fig_trend.write_html("sales_profit_trend.html")
fig_region.write_html("regional_performance.html")
summary_df = df.groupby(['Category', 'Sub-Category'])[['Sales', 'Profit']].sum().reset_index()
summary_df['Margin (%)'] = (summary_df['Profit'] / summary_df['Sales']) * 100
summary_df.to_csv("executive_summary_kpis.csv", index=False)
print("Export complete: Interactive HTML files and CSV summary ready for presentation.")
```

```
Export complete: Interactive HTML files and CSV summary ready for presentation.
```

10. Discussion: Business Value of Data Storytelling: Why storytelling transforms analytical impact.- Data storytelling bridges the gap between raw data analytics and corporate decision-making:

- Contextualizing Metrics: Raw dashboards show what happened; data storytelling explains why it happened and what action to take next.
- Reducing Cognitive Load: Structuring insight sequentially prevents decision paralysis among non-technical executives.
- Driving Alignment: Grounding recommendations in clear, interactive visual evidence accelerates stakeholder buy-in and resource allocation.